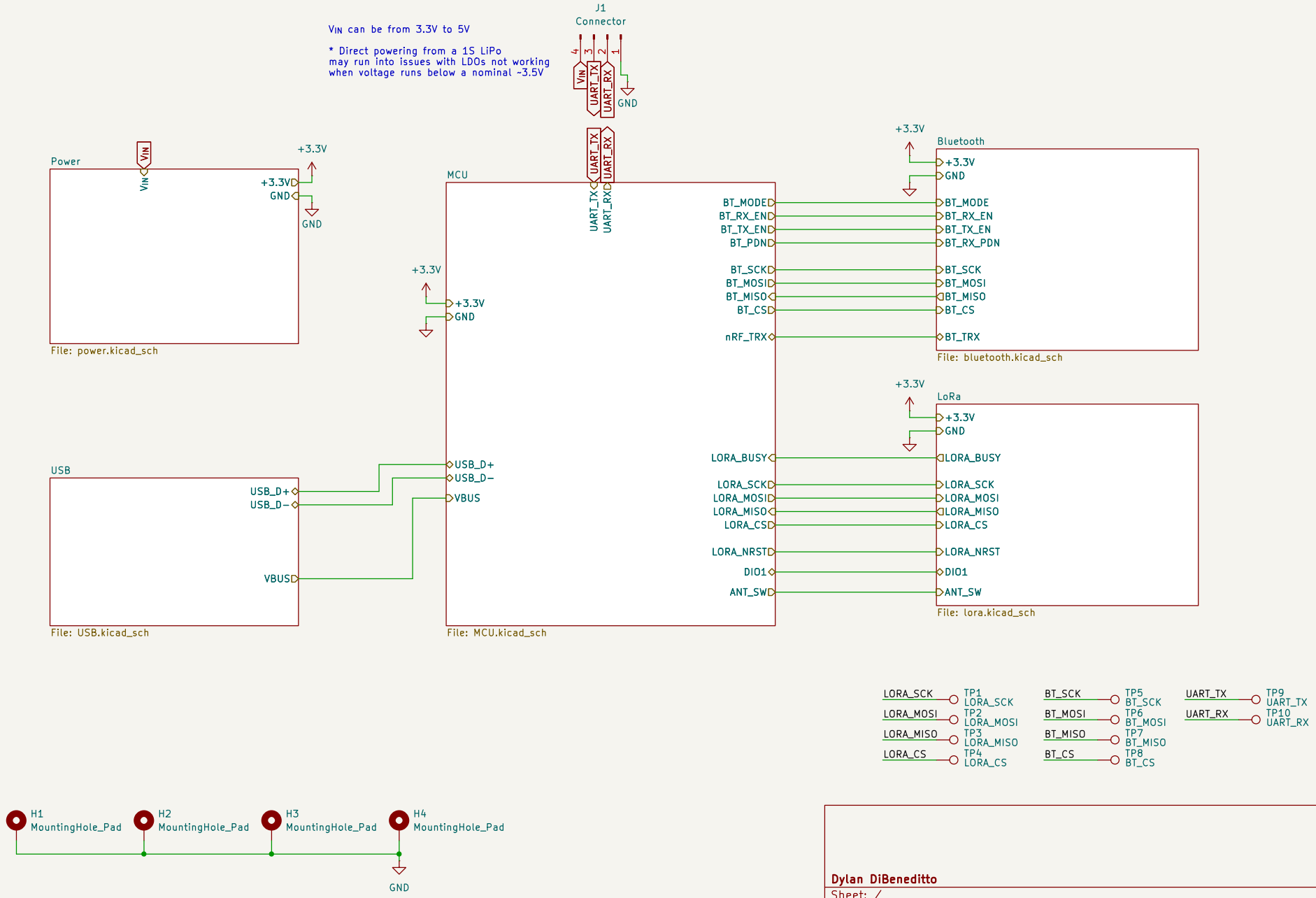
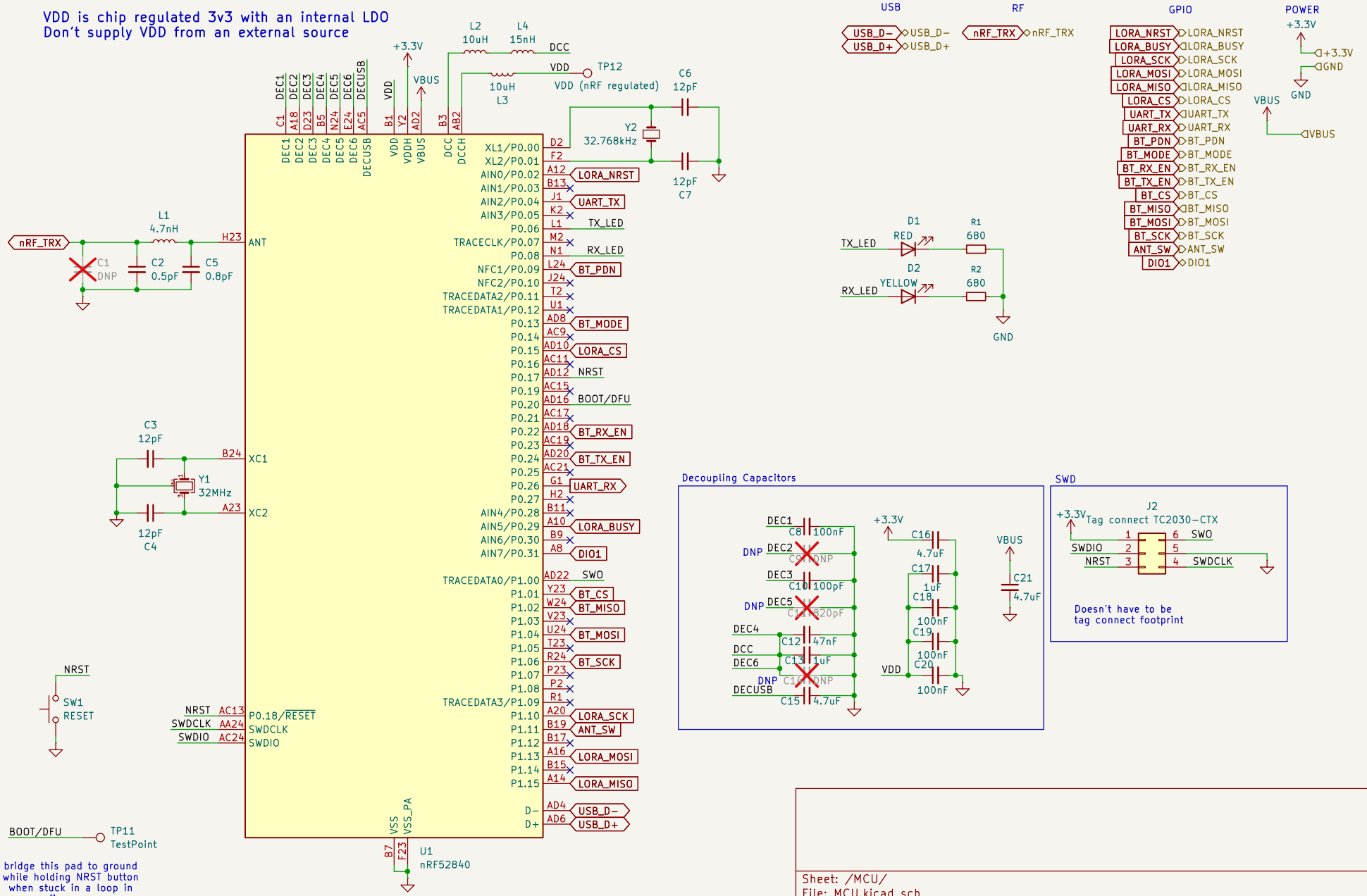
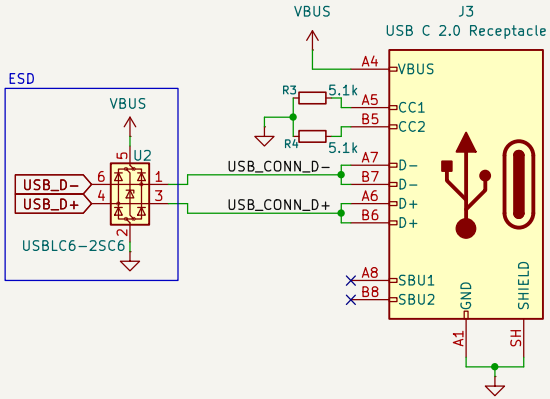
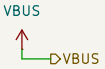

VIN can be from 3.3V to 5V
* Direct powering from a 1S LiPo
may run into issues with LDOs not working
when voltage runs below a nominal ~3.5V
J1
Connector
4
3
2
1
VIN
UART_TX
UART_RX
GND
Power
VIN
+3.3V
GND
File: power.kicad_sch
MCU
UART_TX
UART_RX
+3.3V
GND
BT_MODE
BT_RX_EN
BT_TX_EN
BT_PDN
BT_SCK
BT_MOSI
BT_MISO
BT_CS
nRF_TRX
USB_D+
USB_D−
VBUS
LORA_BUSY
LORA_SCK
LORA_MOSI
LORA_MISO
LORA_CS
LORA_NRST
DIO1
ANT_SW
File: MCU.kicad_sch
Bluetooth
BT_MODE
BT_RX_EN
BT_TX_EN
BT_RX_PDN
BT_SCK
BT_MOSI
BT_MISO
BT_CS
BT_TRX
File: bluetooth.kicad_sch
LoRa
LORA_BUSY
LORA_SCK
LORA_MOSI
LORA_MISO
LORA_CS
LORA_NRST
DIO1
ANT_SW
File: lora.kicad_sch
USB
USB_D+
USB_D−
VBUS
File: USB.kicad_sch
TP1 LORA_SCK
TP2 LORA_MOSI
TP3 LORA_MISO
TP4 LORA_CS
TP5 BT_SCK
TP6 BT_MOSI
TP7 BT_MISO
TP8 BT_CS
TP9 UART_TX
TP10 UART_RX
H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
GND
Dylan DiBeneditto
Sheet: /
File: pcb.kicad_sch
Title: 2.4GHz BT and 915MHz LoRa Transciever
Size: A4
Date: 2026-10-09
Rev: 1
KiCad E.D.A. 10.0.4
Id: 1/6

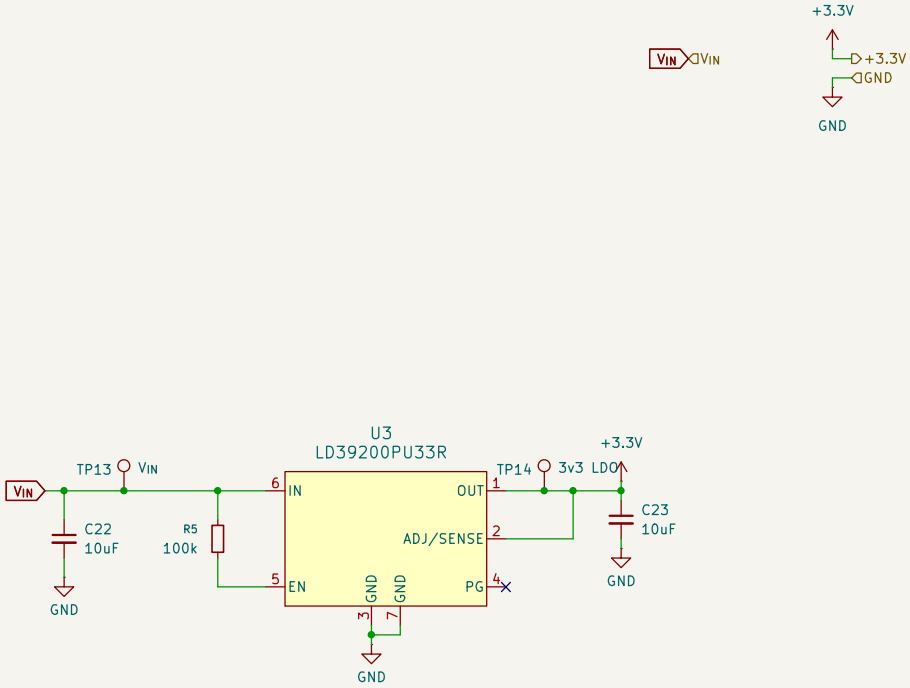

VDD is chip regulated 3v3 with an internal LDO
Don't supply VDD from an external source
USB
USB_D-
USB_D+
RF
nRF_TRX
GPIO
LORA_NRST
LORA_BUSY
LORA_SCK
LORA_MOSI
LORA_MISO
LORA_CS
UART_TX
UART_RX
BT_PDN
BT_MODE
BT_RX_EN
BT_TX_EN
BT_CS
BT_MISO
BT_MOSI
BT_SCK
ANT_SW
DIO1
POWER
+3.3V
GND
VBUS
L1 4.7nH
C1 DNP
C2 0.5pF
C5 0.8pF
L2 10uH
L4 15nH
L3 10uH
DCC
VDD
TP12
VDD (nRF regulated)
C6 12pF
Y2 32.768kHz
C7 12pF
C3 12pF
Y1 32MHz
C4 12pF
D1 RED
R1 680
D2 YELLOW
R2 680
TX_LED
RX_LED
XL1/P0.00
XL2/P0.01
AIN0/P0.02
AIN1/P0.03
AIN2/P0.04
AIN3/P0.05
P0.06
TRACECLK/P0.07
P0.08
NFC1/P0.09
NFC2/P0.10
TRACEDATA2/P0.11
TRACEDATA1/P0.12
P0.13
P0.14
P0.15
P0.16
P0.17
P0.19
P0.20
P0.21
P0.22
P0.23
P0.24
P0.25
P0.26
P0.27
AIN4/P0.28
AIN5/P0.29
AIN6/P0.30
AIN7/P0.31
TRACEDATA0/P1.00
P1.01
P1.02
P1.03
P1.04
P1.05
P1.06
P1.07
P1.08
TRACEDATA3/P1.09
P1.10
P1.11
P1.12
P1.13
P1.14
P1.15
D-
D+
ANT
XC1
XC2
P0.18/RESET
SWDCLK
SWDIO
VSS
VSS_PA
U1
nRF52840
NRST
SW1
RESET
BOOT/DFU
TP11
TestPoint
bridge this pad to ground while holding NRST button when stuck in a loop in firmware
Decoupling Capacitors
C8 100nF
C9 DNP
C10 100pF
C11 820pF
C12 47nF
C13 1uF
C14 DNP
C15 4.7uF
C16 4.7uF
C17 1uF
C18 100nF
C19 100nF
C20 100nF
C21 4.7uF
SWD
J2
Tag connect TC2030-CTX
SWO
SWDCLK
Doesn't have to be tag connect footprint
Sheet: /MCU/
File: MCU.kicad_sch
Title:
Size: A4
Date: 2026-10-09
Rev:
KiCad E.D.A. 10.0.4
Id: 2/6



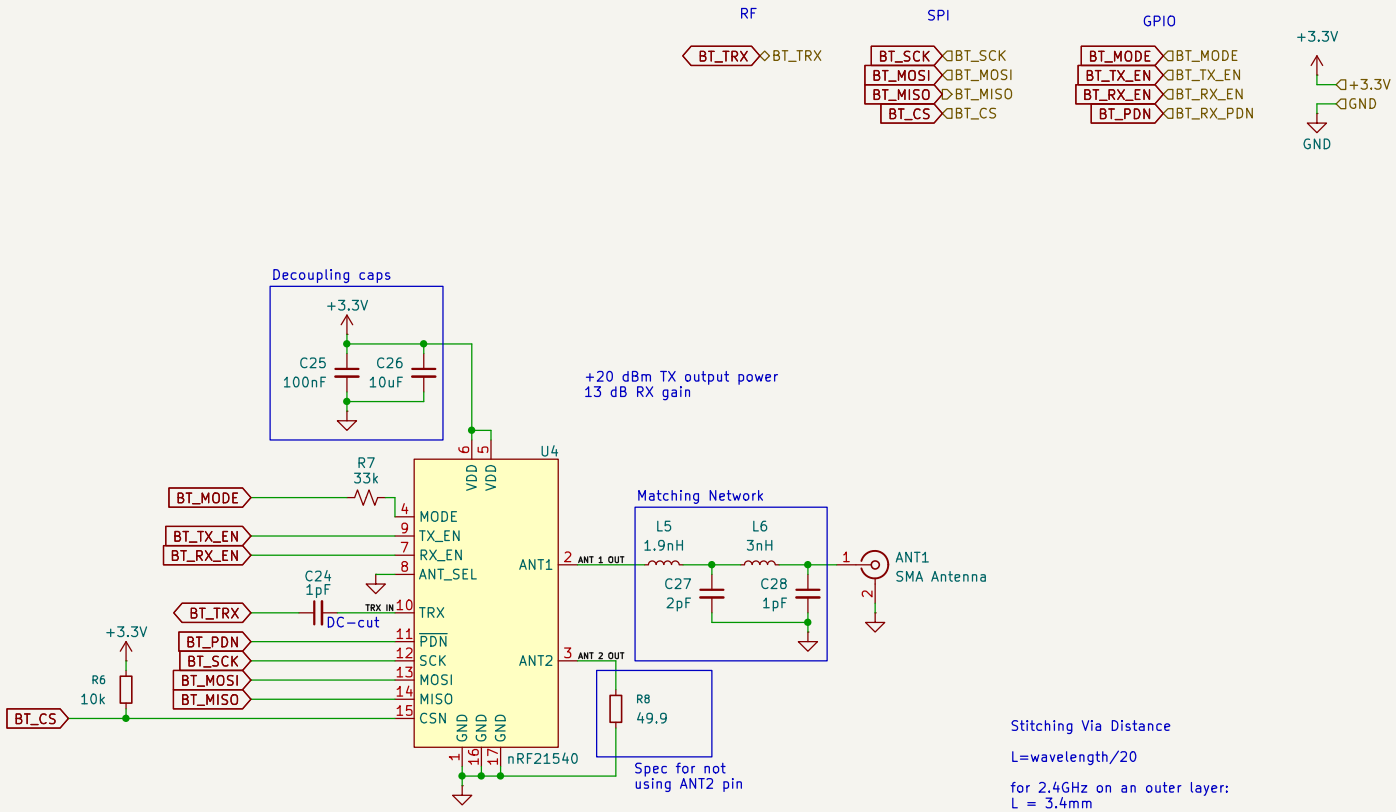




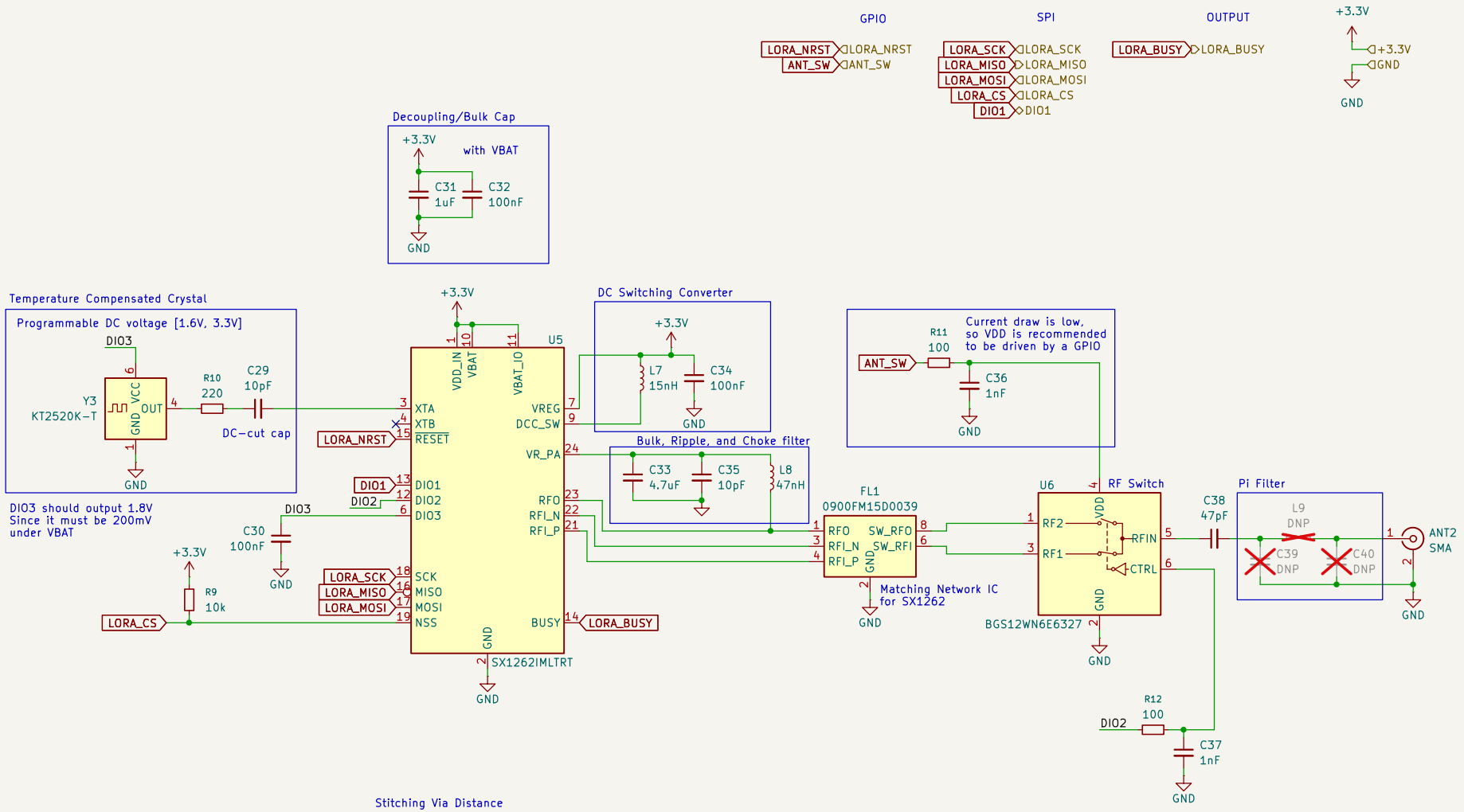
Dylan DiBeneditto		
Sheet: /USB/ File: USB.kicad_sch		
Title: USB−C 2.0 and ESD		
Size: A4	Date: 2026−10−09	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/6



Dylan DiBeneditto		
Sheet: /Power/ File: power.kicad_sch		
Title: Power		
Size: A4	Date: 2026-10-09	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/6



Dylan DiBeneditto		
Sheet: /Bluetooth/		
File: bluetooth.kicad_sch		
Title: Bluetooth		
Size: A4	Date: 2026-10-09	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/6



Dylan DiBeneditto		
Sheet: /LoRa/		
File: lora.kicad_sch		
Title: 915MHz LoRa Modem		
Size: A4	Date: 2026-10-09	**Rev: 1**
KiCad E.D.A. 10.0.4		Id: 6/6